Survival Analysis of Rats Data

A Comparative Study of Parametric and Regression Models

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github.com/walternyamutamba-svg/rats-survival-analysis

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Objective

Goal

Fit six parametric survival distributions to the `rats` tumour-incidence data, select the best model using AIC, and derive the four life functions: $S(t)$, $h(t)$, $f(t)$, $H(t)$.

Why Parametric Models?

- Kaplan-Meier is non-parametric which means it has no closed-form expression
- Parametric models yield analytic formulas for survival, hazard, density, and cumulative hazard
- Enable direct computation of mean, median, and variance of survival time
- Allow extrapolation beyond the observed follow-up period

Dataset & Study Design

Source: `rats` dataset, `survival` package in R

Variables used:

- `time` — days to tumour or censoring
- `event` — 1 = tumour, 0 = censored
- `treatment` — Control vs Treated (rx)
- `sex` — female / male covariate
- `litter` — clustering variable (100 litters)

Design: 2:1 allocation
200 Control, 100 Treated
Max follow-up = 104 days

Treated rats received a carcinogenic agent (rx = 1)

Table 1: Sample Summary

Quantity	Value
Total rats	300
Events (tumours)	42
Censored	258
Overall event rate	14%
Control events	21/200 (10.5%)
Treated events	21/100 (21.0%)
Mean survival	90.4 days
Missing values	None

Note: Low event rate (14%) means median survival is not estimable so we report RMST

Descriptive Overview

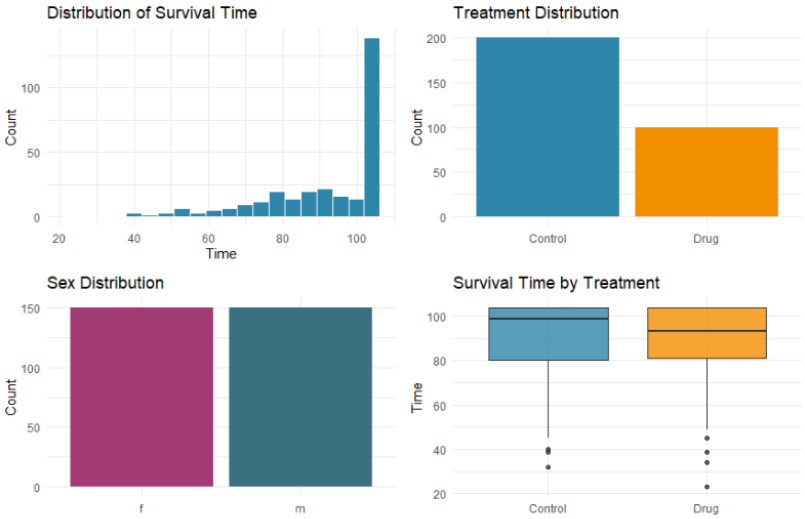

Figure 1: Distribution of survival times and comparison by treatment group

Kaplan–Meier Survival Curves

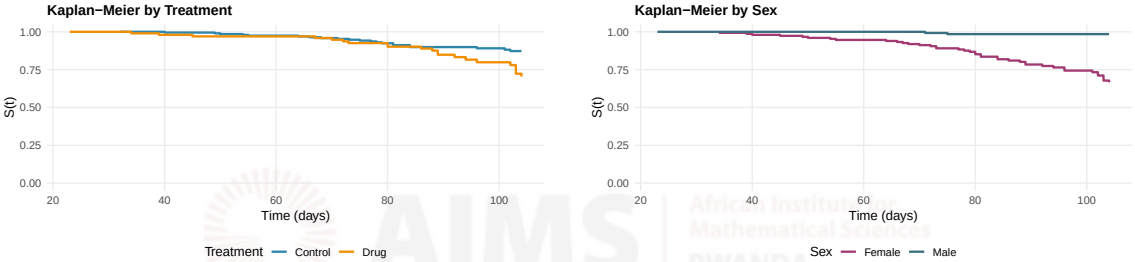

Figure 2: Kaplan–Meier survival curves

Interpretation

- Females experience substantially more tumour events than males (40 vs 2).
- The treatment effect appears modest, with similar survival patterns between groups.
- Kaplan–Meier curves provide a non-parametric reference for evaluating parametric survival models.

Parametric Model Fitting and Selection

Table 2: Parametric Model Values

Model	Params	Estimates	AIC	loglikelihood
Exp	λ	0.00155	629.54	-313.77
Weibull	k, λ	3.67,160.67	**578.20**	-287.10
LogN	μ, σ	5.16,0.56	578.86	-287.43
LL	α, β	153.95,3.82	578.32	-287.16
Gamma	k, λ	5.48,0.032	578.38	-287.19
Gomp	b, η	0.0001,0.044	580.66	-288.33

Why AIC?

- Balances goodness of fit and model complexity
- Penalises models with too many parameters
- Used to compare non-nested survival models
- Lower AIC indicates a better model

Comparison — All Models vs KM

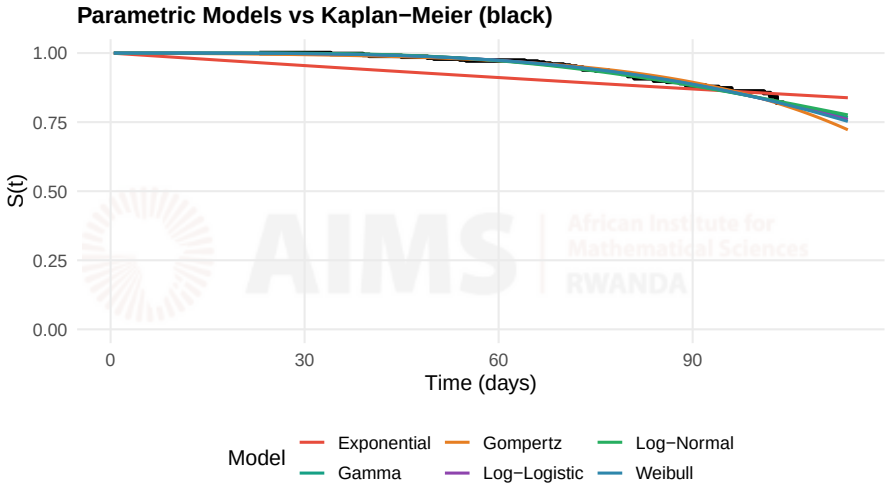

Figure 3: Comparison of parametric survival models with the Kaplan–Meier curve (black).

Why Weibull?

Statistical Justification

- Lowest AIC (578.2) among all six candidates
- Closest visual alignment with non-parametric KM curve
- Includes Exponential as a special case ($k = 1$), so rejecting Exponential supports $k \neq 1$

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Practical Justification

- Shape parameter k directly characterises the hazard trend
- Closed-form expressions for all four life functions
- Mean, median, and variance available analytically
- Widely used and interpretable in biomedical research

Weibull Life Functions

With shape $k = 3.67$ and scale $\lambda = 160.67$

Survival function

$$S(t) = \exp\left[-\left(\frac{t}{\lambda}\right)^k\right]$$

Hazard function

$$h(t) = \frac{k}{\lambda}\left(\frac{t}{\lambda}\right)^{k-1}$$

Density function

$$f(t) = h(t) \cdot S(t)$$

Cumulative hazard

$$H(t) = \left(\frac{t}{\lambda}\right)^k$$

Since $k = 3.67 > 1$, the hazard $h(t)$ is monotonically increasing which means tumour risk accelerates over time.

Weibull Life Functions

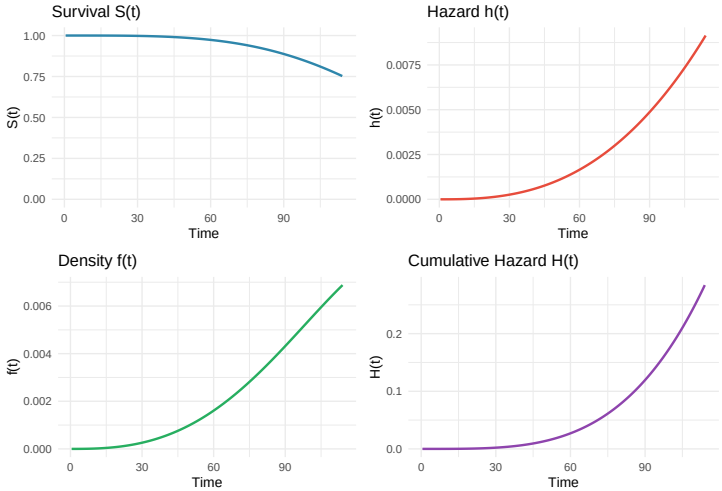

Figure 4: Weibull life functions showing survival $S(t)$, hazard $h(t)$, density $f(t)$, and cumulative hazard $H(t)$.

Interpretation

Parameter Meanings

- $k = 3.67$ hazard is increasing and convex which means tumour risk accelerates with age
- $\lambda = 160.67$ the characteristic life (time at which $S(t) = e^{-1} \approx 0.37$)

Summary Statistics

Table 3: Summary Statistics

Quantity	**Value**
Mean survival	144.9 days
Median survival	145.4 days
Variance	1 934.8 days^2
Std. deviation	44.0 days

Conclusions

1. The Weibull distribution provides the best fit (AIC $= 578.2$), closely followed by Log-Logistic and Gamma
2. The shape $k = 3.67 > 1$ confirms an increasing hazard which meant that tumour risk grows with time
3. Mean tumour-free survival is approximately 145 days
4. The Exponential model (constant hazard) is decisively rejected (ΔAIC $= 51.3$), justifying the need for a flexible parametric model

Limitations

- Rats from the same litter may be similar. A shared frailty model can account for this clustering.
- Treatment and sex were not included in the parametric models.

References

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